

Residual stress in the D2C metric solve a reference metric that fluctuates on the patch scale

Setup

Per triangle s , split the *actual* and *reference* metrics into a macroscopic mean plus a fluctuation:

$$g_s = g + \delta g_s, \quad \bar{g}_s = \bar{g} + \delta \bar{g}_s, \quad \Delta g \equiv g - \bar{g}.$$

The new ingredient is $\delta \bar{g}_s$: the reference metric now varies *within* a patch, not only the actual metric. The fluctuation is *not* a free field — it is fixed by the response ansatz

$$\delta g_s = W_s \Delta g,$$

and the variational unknown of the forward solve is the loading-independent operator W_s . The energy is minimised **with respect to** W_s (subject to the compatibility constraints), not with respect to δg_s . This is the point I first got wrong.

What does *not* change. The edge carrier q_e is a coordinate vector (lab geometry); it is unchanged. The compatibility constraints (edge / angle–curvature / area–mean) encode *flat-space achievability of the actual network* — the nodes really sit in the plane — and are unchanged. The reference $\bar{g}_s$ need *not* be flat-embeddable; its incompatibility does **not** enter as a constraint right-hand side.

Regime 1 — $\bar{g}$ slowly varying (scale $\gg$ patch)

Over one patch $\bar{g}$ is essentially constant, so $\delta \bar{g}_s \approx 0$. Then $\bar{g}$ is merely a relabelled constant reference: $\Delta g = g - \bar{g}$ is generically nonzero, but the local problem is the old one to leading order,

$$\delta g_s = W_s \Delta g, \quad W_s \text{ exactly as before.}$$

Nothing to change — “just rename the reference metric.”

Regime 2 — $\bar{g}$ varying on the patch scale ($\delta \bar{g}_s \neq 0$)

Energy. Write the local mismatch as $r_s \equiv (I + W_s)\Delta g - \delta \bar{g}_s$. Then

$$E = \sum_s A_s (g_s - \bar{g}_s)^2 = \sum_s A_s (\Delta g + \delta g_s - \delta \bar{g}_s)^2 = \sum_s r_s^\top A_s r_s, \quad A_s(\cdot)^2 \equiv (\cdot)^\top A_s(\cdot),$$

with A_s and q_e built from the same coordinate geometry as before.

Stationarity (matched to the existing solve). The variational unknown is the loading-independent operator W_s , with the field fixed by $\delta g_s = W_s \Delta g$; minimising over W_s is equivalent to minimising over the field δg_s and then stripping Δg (everything is linear in it), which is how the existing solve is written (INTRINSIC_METRIC_SOLVE.md §3.3). In the metric two-index notation $g(s)_{\mu\nu}$, with four-index $A(s)^{\mu\nu\alpha\beta}$, four-index $W(s)_{\mu\nu}^{\alpha\beta}$ (9 components), and $\delta g(s)_{\mu\nu} = W(s)_{\mu\nu}^{\alpha\beta} \Delta g_{\alpha\beta}$, the per-triangle stationarity is

$$A(s)^{\mu\nu\alpha\beta} (\Delta g + \delta g(s) - \delta \bar{g}(s))_{\alpha\beta} + [\lambda_e, \kappa_v, \chi \text{ terms}]^{\mu\nu} = 0, \quad \delta g(s) = W(s) \Delta g,$$

with the *single* free index pair $\mu\nu$ and the bracket the edge (λ_e) + curvature (κ_v) + area-mean (χ) multiplier terms. Setting $\delta \bar{g} = 0$ is exactly the existing boxed equation $A(s)^{\mu\nu\alpha\beta} (\Delta g + \delta g(s))_{\alpha\beta} + [\dots]^{\mu\nu} = 0$; stripping Δg then gives the loading-independent operator solve with right-hand side $-A$ (INTRINSIC_METRIC_SOLVE.md Eq. after the Lagrangian). The *only* new element is the substitution $(\Delta g + \delta g) \rightarrow (\Delta g + \delta g - \delta \bar{g})$ inside the mismatch. (Sign: $-\delta \bar{g}$ from the energy; a $+\delta \bar{g}$ convention flips it.)

Why not a trailing Δg . Differentiating the energy w.r.t. W (rather than the field) attaches an extra factor $\partial \delta g / \partial W = \Delta g$, giving $f_s \otimes \Delta g$ with f_s the bracket above. The key point is that the *same* Δg multiplies the constraint variation: the three constraints are imposed on the **field** $\delta g = W \Delta g$ — edge $(J \delta g)_e = 0$, curvature $(C \delta g)_v = 0$, mean $M_S \delta g = 0$ (INTRINSIC_METRIC_SOLVE.md §3) — so each carries Δg , and its W -variation carries the identical trailing $\Delta g_{\rho\sigma}$ with the same free indices. It factors out, leaving $f_s = 0$ — the force-balance form above, free indices $\mu\nu$ only, not $\mu\nu\alpha\beta$. (A term would survive *only* if a constraint were operator-level, on W with no Δg ; none is.) Carrying the trailing Δg would over-index the equation relative to the existing one.

What the reference fluctuation does. $\delta g(s) = W(s) \Delta g$ holds *always*. The reference fluctuation $\delta \bar{g}(s)$ enters only through the mismatch $(\Delta g + \delta g - \delta \bar{g})$, i.e. as the **added source** $A(s)^{\mu\nu\alpha\beta} \delta \bar{g}(s)_{\alpha\beta}$ on the right-hand side of the *same* block system: it shifts the target the mismatch is measured against, so the solved response acquires a $\delta \bar{g}$ -driven correction. The operator $A(s)$, the carriers q_e , and the constraints (edge / curvature / mean) are untouched. (Note: once $\delta \bar{g} \neq 0$ the source is no longer purely $\propto \Delta g$, so Δg cannot be cleanly stripped as at $\delta \bar{g} = 0$ — the precise consequence of that, and the self-consistent definition of $\delta \bar{g}$, are deferred.)

Consequences

- **q_e , A_s , and the constraints are unchanged**, and W is still obtained from the same KKT system — now with the extra source $A(s)^{\mu\nu\alpha\beta} \delta \bar{g}(s)_{\alpha\beta}$ on the right-hand side.
- **Cheap and differentiable.** It is an added source term (plus the mismatch/pre-stress readout), not new operators; the existing factorisation and its gradients are reused.
- **Reduces correctly.** As $\delta \bar{g}_s \rightarrow 0$ (slowly varying reference) everything collapses to the current framework.
- **Residual / pre-stress** lives in the mismatch $\sigma_s = A_s r_s = A_s [(I + W_s) \Delta g - \delta \bar{g}_s]$, whose $-A_s \delta \bar{g}_s$ piece is the reference-frustration contribution. Exactly how the zero-macroscopic-load residual stress is extracted within the $\delta g_s = W_s \Delta g$ ansatz is left open here (see below), not asserted.

The genuine open problem

Everything above assumes $\delta\bar{g}_s$ is *given*. The real work is defining it *self-consistently*: what counts as “small scale,” and how to split a prescribed reference field into the patch-mean $\bar{g}$ (absorbed into Δg , Regime 1) versus the sub-patch fluctuation $\delta\bar{g}_s$ (the source, Regime 2). That scale separation — a self-consistent definition of the patch reference metric — is the conceptual crux, not the solver mechanics.